

WASP-52b

R-1433

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_211005 · created 2026-08-01T07:43:17+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459492.7303 +/- 0.0064 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.055
Transit depth (Rp/Rs)^2	2.34 +/- 1.68 [%]
Orbital Inclination (inc)	89.95 +/- 2.94
Airmass coefficient 1 (a1)	0.734 +/- 0.057
Airmass coefficient 2 (a2)	0.25 +/- 0.13
Scatter in the residuals of the lightcurve fit is	7.61 %
Best Comparison Star	#3 - [462, 167]
Optimal Aperture	4.98
Optimal Annulus	23.66
Transit Duration (day)	0.0825 +/- 0.0088

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.153 ± 0.055 vs 0.1646 (deviation 0.17σ)
Duration fit vs published	0.0825 d vs 0.0758 d (deviation 0.63σ)
Mid-time O–C	16.3 min
Depth SNR	2.3
Residual scatter	7.61 %

Light curve

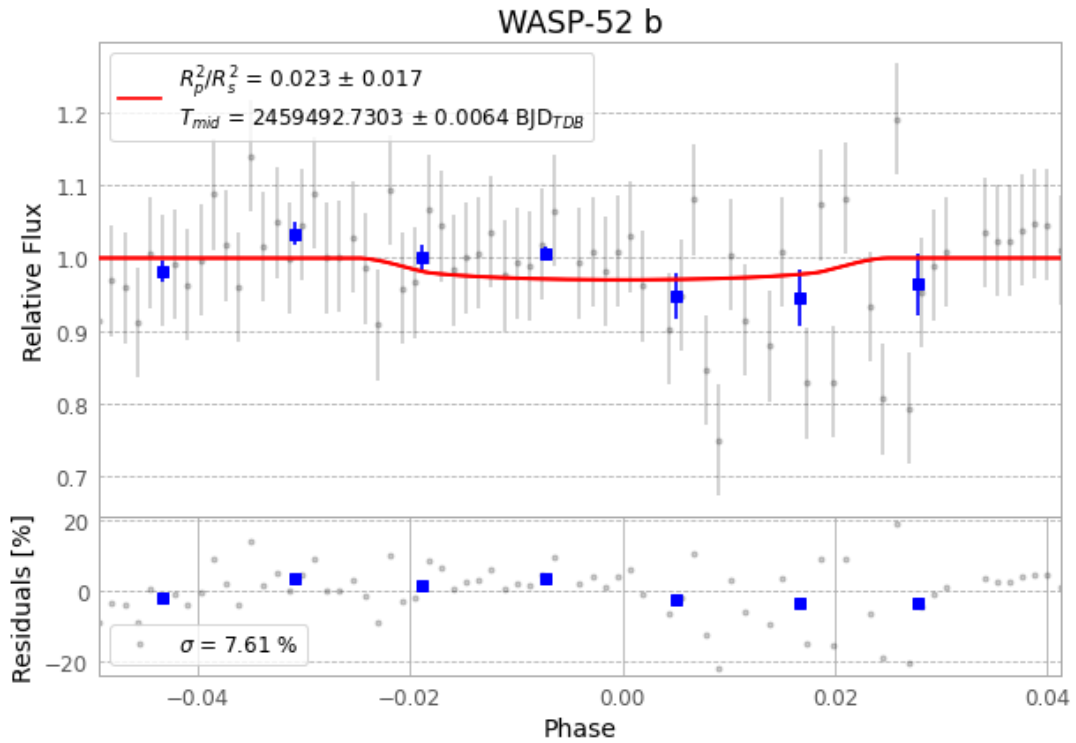


Figure 1 — FinalLightCurve_WASP-52 b_2021-10-04.png (SHA-256 407b9cbcd8546e54...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [385, 111], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6009af1b6bde8175...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), WASP-52211005031812.FITS → WASP-52211005070612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-211005.log (ad3e8985ec0f...), FinalLightCurve_WASP-52 b_2021-10-04.png (407b9cbcd854...), FinalLightCurve_WASP-52 b_2021-10-04.csv (66957da3d432...)

Compute resources

Wall time 180.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 07:43Z.